Objective

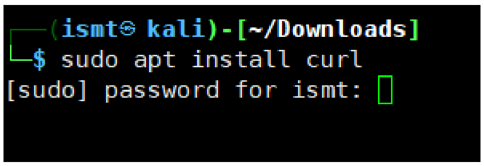
Splunk configuration to understand basic SIEM function

Linux Server

1. Keep Kali Linux system updated.

   $ sudo apt –y update
   $ sudo apt –y upgrade

2. Install Curl Package in Kali Linux which is required by splunk

3. Download Trail version of Splunk Enterprise

   From Official Website:

   https://www.splunk.com/

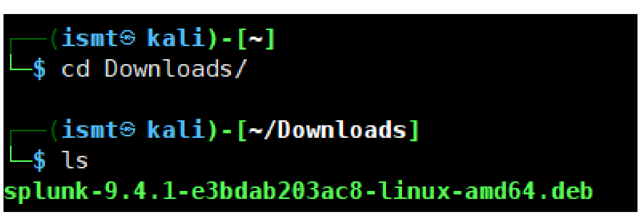
   Or,

   Google Drive download link:

   https://drive.google.com/drive/folders/1h9ZevccIrflTuMVxGy94m2pIVJ8dwCva?usp=drive_link

4. Copy downloaded Splunk package to Downloads folder of Kali Linux. Verify it using following commands

   $ cd Downloads/
   $ ls

5. Install the splunk package in Kali Linux

```
┌──(ismt㉿ kali)-[~/Downloads]
└─$ sudo dpkg -i splunk-9.4.1-e3bdab203ac8-linux-amd64.deb
[sudo] password for ismt:
```

6. After installation is complete, use following command to accept the license agreement.

   $ sudo /opt/splunk/bin/splunk start --accept-license --answer-yes

```
┌──(ismt㉿ kali)-[~/Downloads]
└─$ sudo /opt/splunk/bin/splunk start --accept-license --answer-yes
```

   Above command will prompt for administrative username and password setup

```
┌──(ismt㉿ kali)-[~/Downloads]
└─$ sudo /opt/splunk/bin/splunk start --accept-license --answer-yes

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username:
```

   For this lab, following username and password is provided:

   Username: ismt
   Password: Ismt@12345

   Once the setup is completed, login URL is also displayed.

   Login URL: http://127.0.0.1:8000

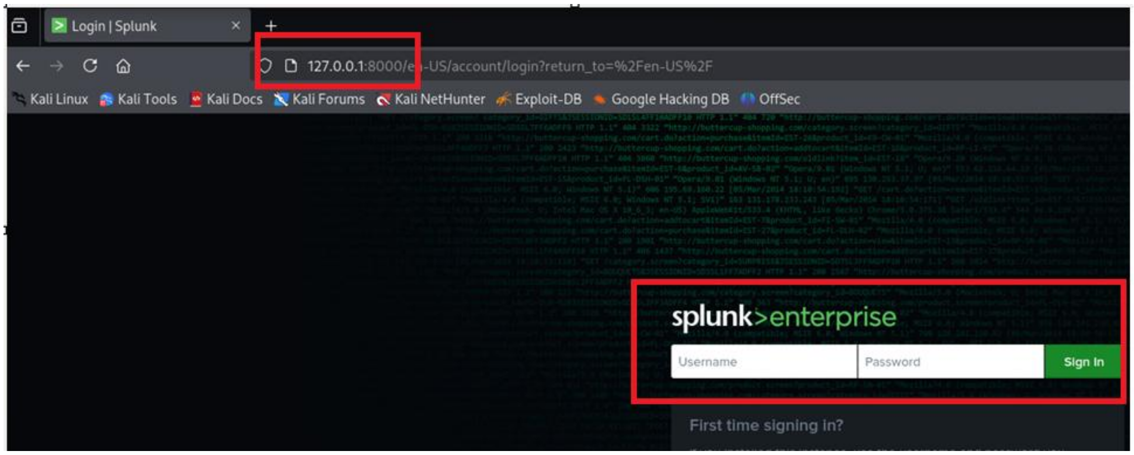
```
Waiting for web server at http://127.0.0.1:8000 to be available.............. Done
```

7. Start splunk service to access the URL

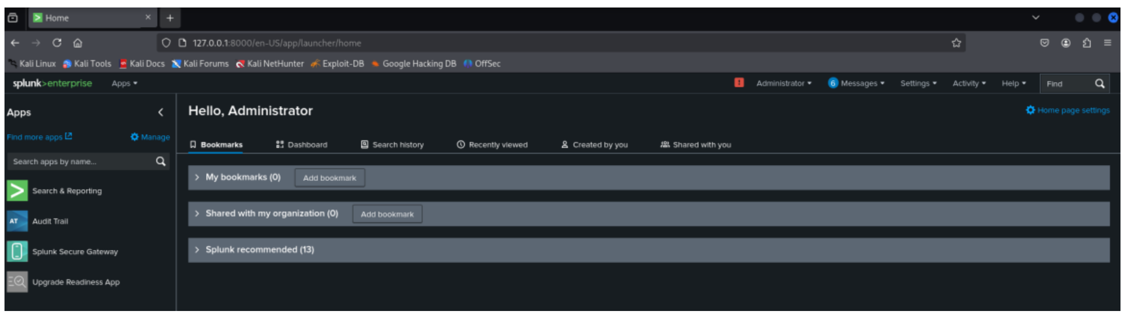
   $ sudo /opt/splunk/bin/splunk start

```
┌──(ismt㉿ kali)-[~/Downloads]
└─$ sudo /opt/splunk/bin/splunk start
```

8. Check if the Splunk web URL is accessible via Browser in Kali Linux

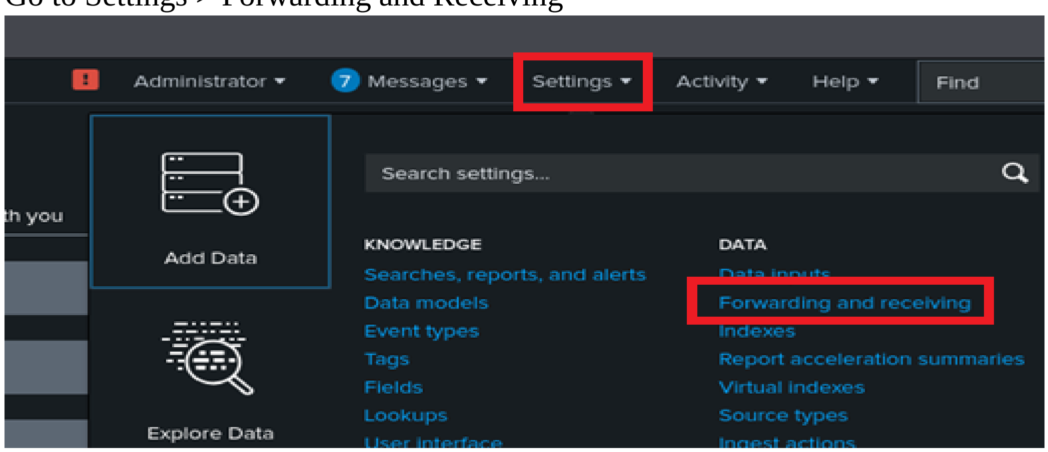


9. Login to the Web Portal using Username and Password provided in Step No – 6.

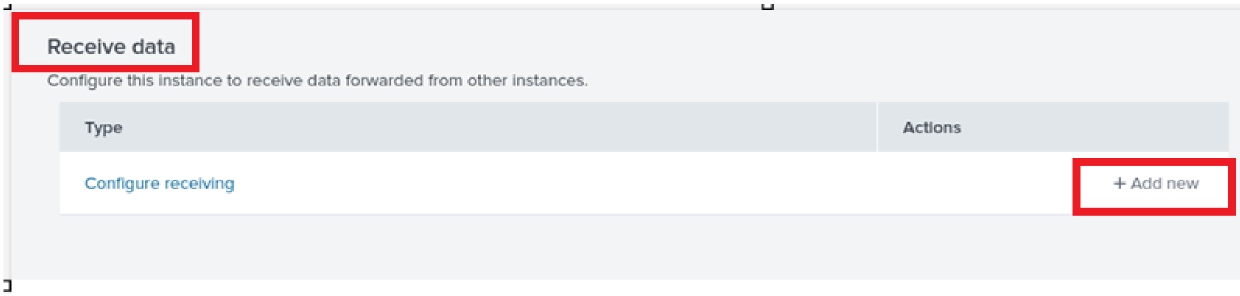


10. Since Splunk application is a SIEM solution, it should be configured to receive/accept logs from other assets.

Go to Settings > Forwarding and Receiving



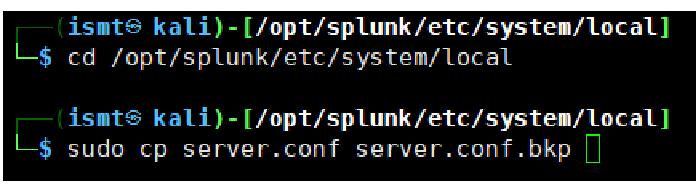
Under receive data section click on Add New



Provide Listening Port as **“9997”** and save it

11. We are configuring splunk in virtual environment where the storage is low, hence we need to make some adjustment in splunk configuration file.

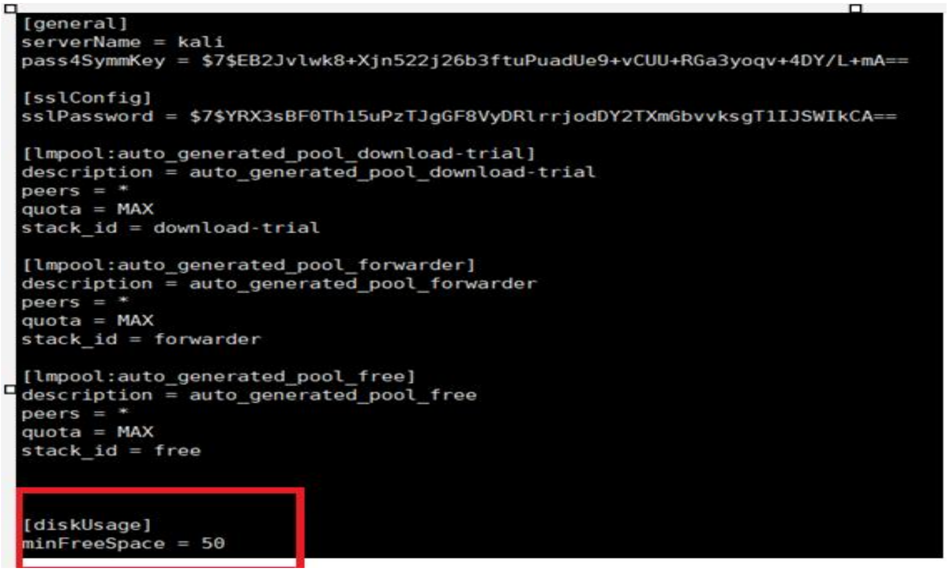
$ cd /opt/splunk/etc/system/local
$ sudo cp server.conf server.conf.bkp



$ sudo vi server.conf

Add the following at the end of the configuration file and save it.

[diskUsage]
minFreeSpace = 50



Verify if the configuration is properly saved.

$ sudo cat server.conf

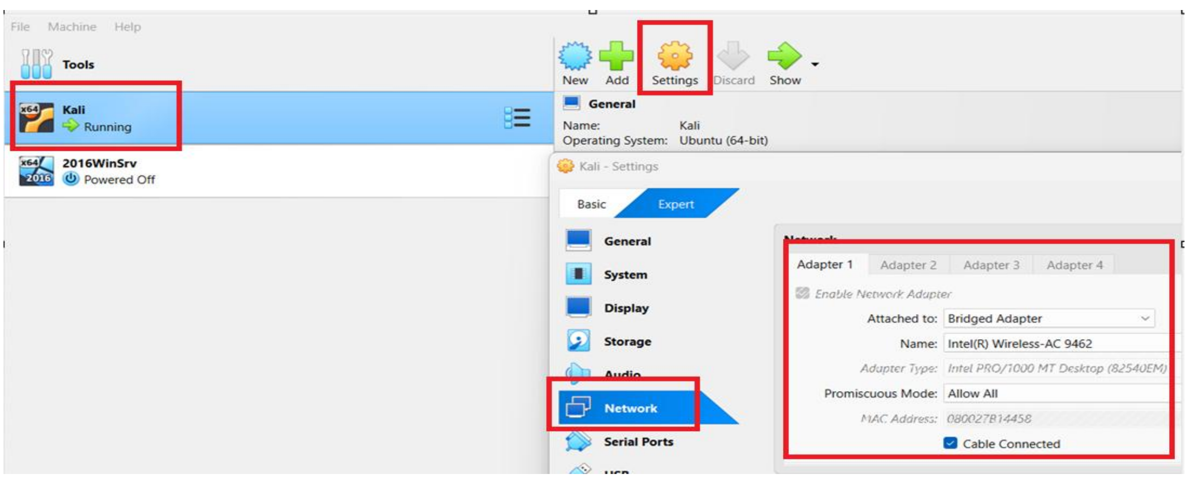
12. Stop and Start the service for configuration to reload

$ sudo /opt/splunk/bin/splunk stop
$ sudo /opt/splunk/bin/splunk start

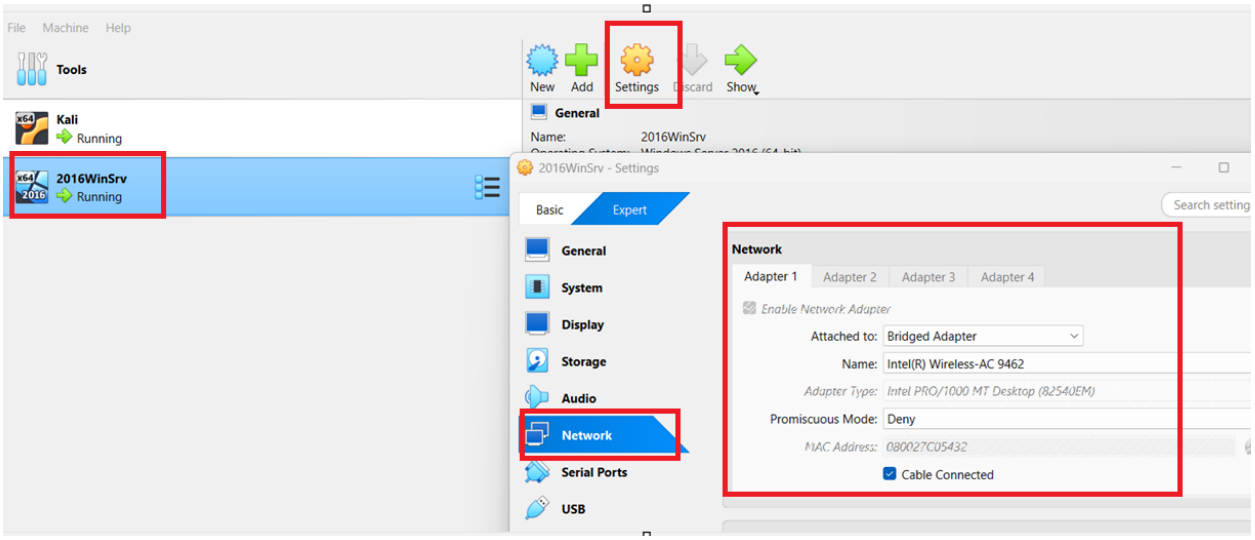
<u>Virtual Box</u>

For this Lab Purpose let us configure different Virtual Box Network for smooth operation.

1. Select Kali Linux Box > Click on Setting > Network

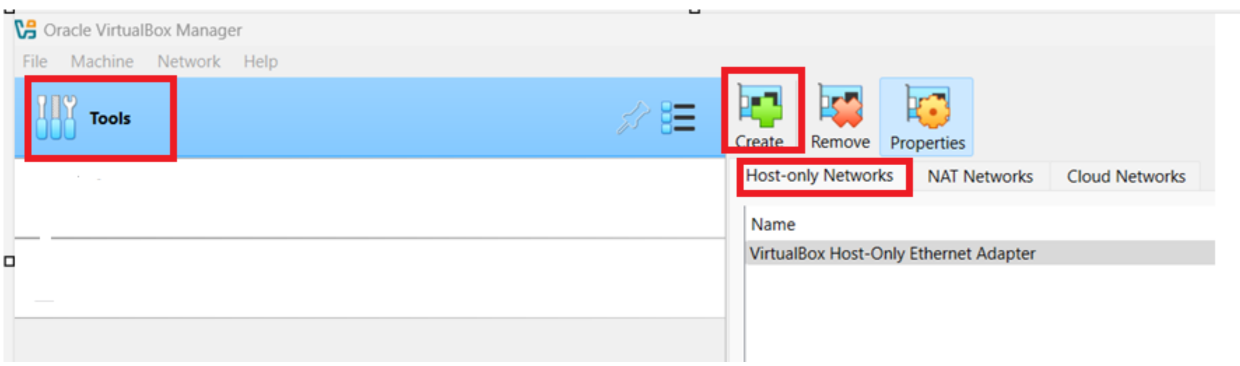


Select Windows Server Box > Click on Settings > Network



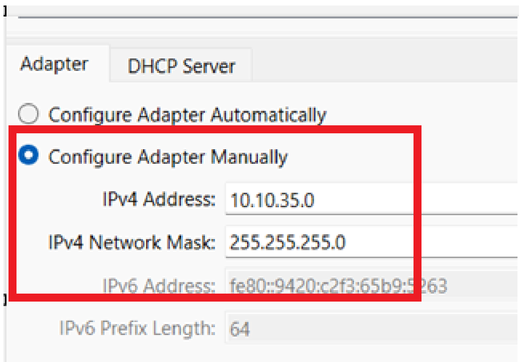
Note: Please keep this screenshot safe as we have to revert back the setting to this state after completion of this Lab.

2. Go to Tools > Select Host-Only Networks and Click on Create

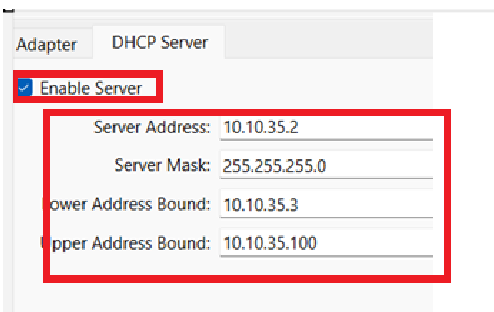


New VirtualBox Host-Only Adaptor will be created.

Change the Adapter IP address. For this LAB, “**10.10.35.0/24**” is used.



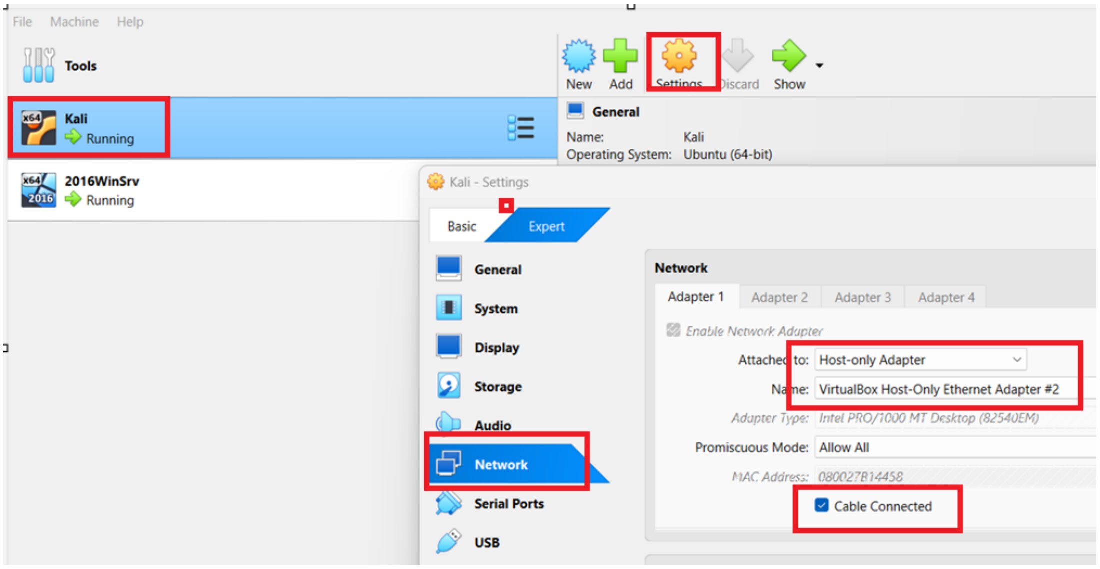
Enable DHCP server for this adaptor and set up DHCP parameters as shown

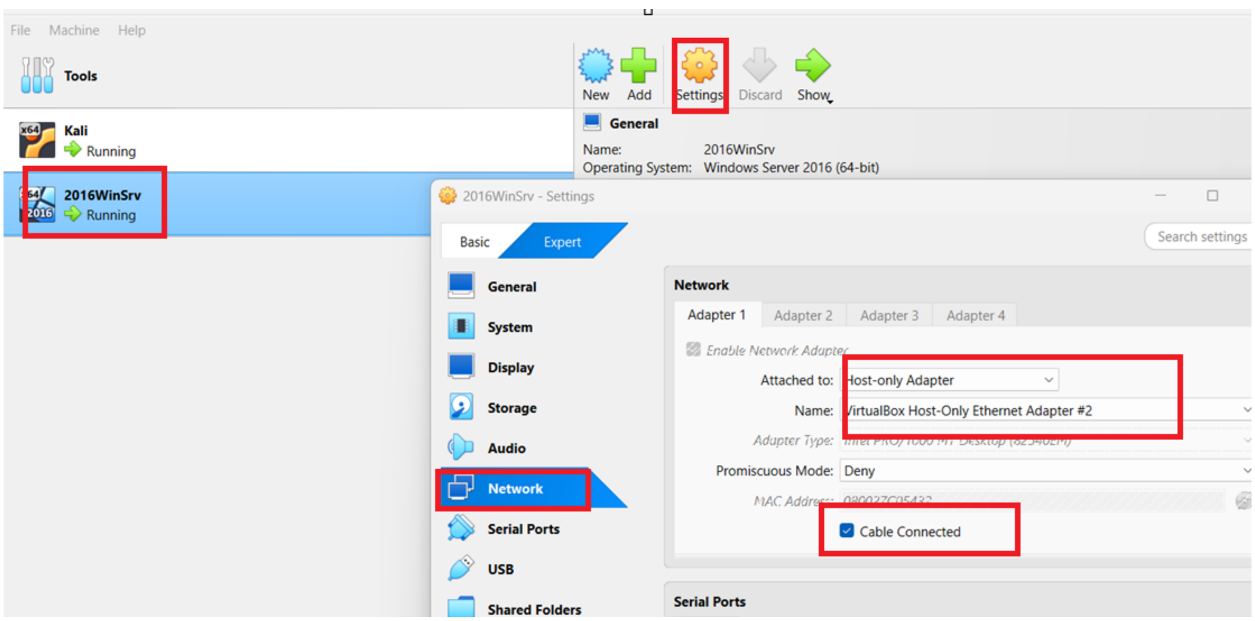


Click on Apply

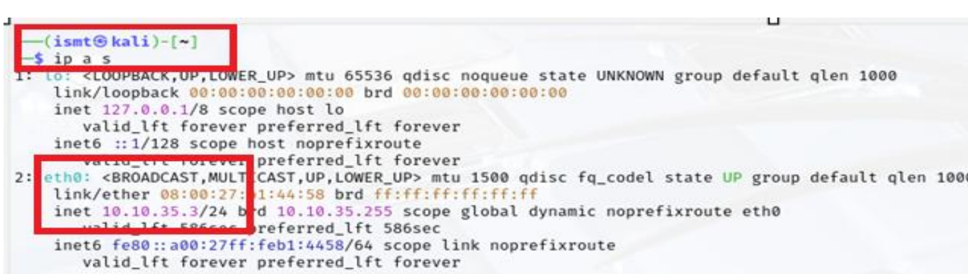
Finally, New VirtualBox Host Only Adaptor will be created.

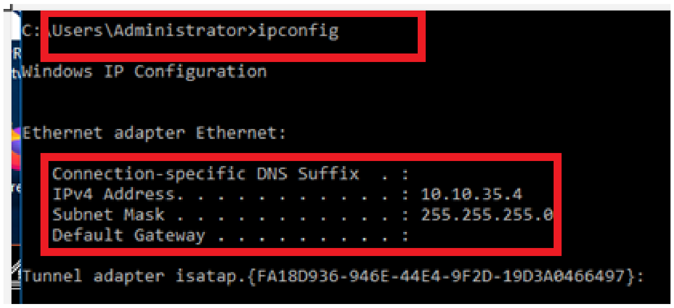
3. Assign this Adaptor to Kali Linux and Windows Server.





4. Check the IP Address received by Kali Linux and Windows Server.





In this Lab, Following IP Address are assigned via DHCP.

Kali Linux – 10.10.35.3
Window Server – 10.10.35.4

Note: These IP are essential for this LAB, please make note of these IPs.

Windows Server

1. Download Splunk Universal Forwarder

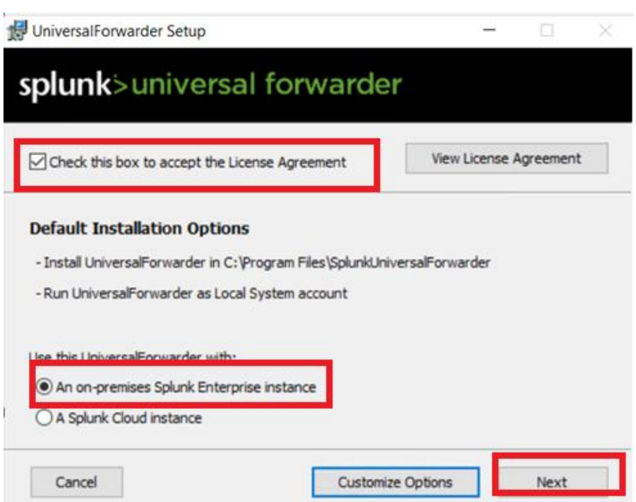
   Form Official Website:
   https://www.splunk.com/

   Or,

   From Google Drive Link:

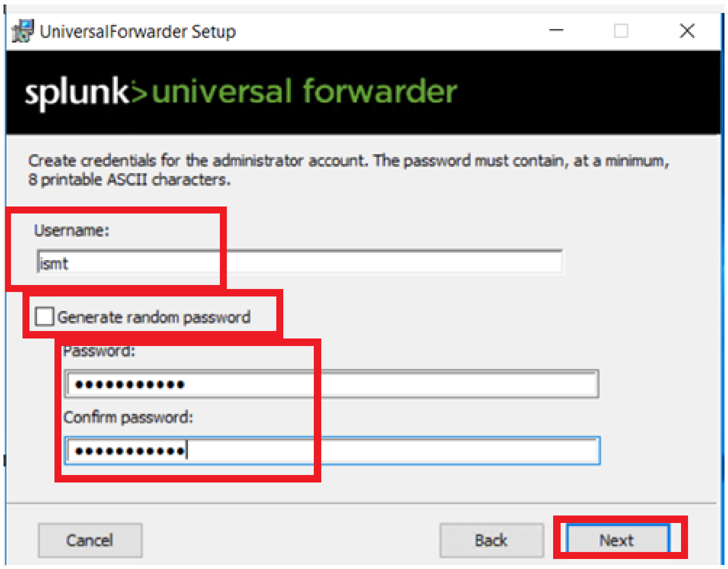
   https://drive.google.com/drive/folders/1h9ZevccIrflTuMVxGy94m2pIVJ8dwCva?usp=drive_link

2. Double Click Splunk Universal Forwarder in Windows Server.

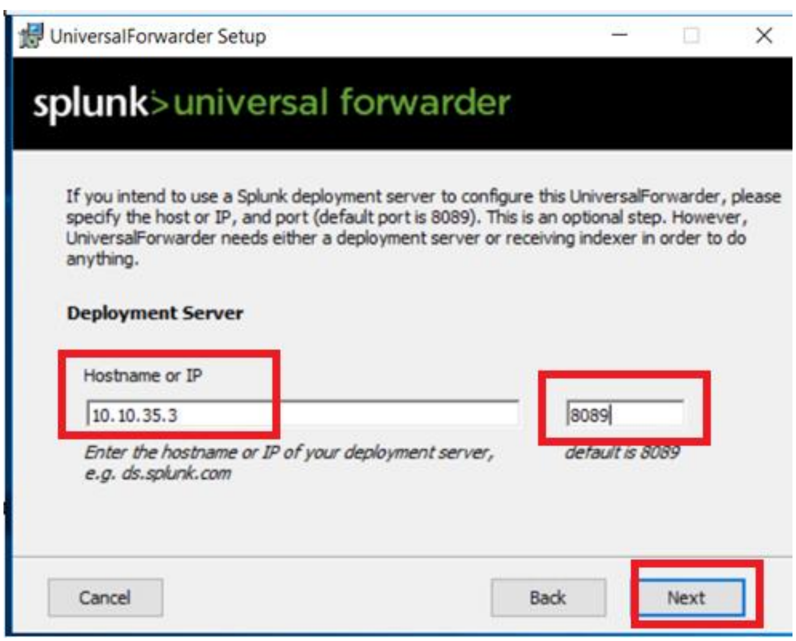


3. Prompt to create administrative user will be provided. For this Lab following details provided.

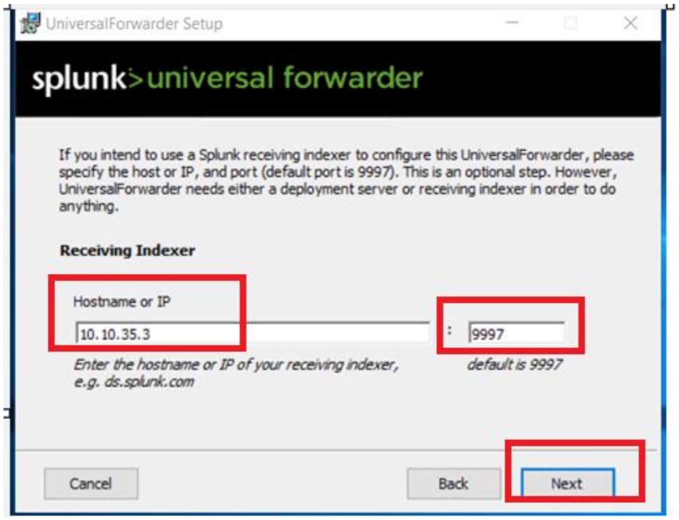
   Username: ismt
   Password: Ismtlab@123



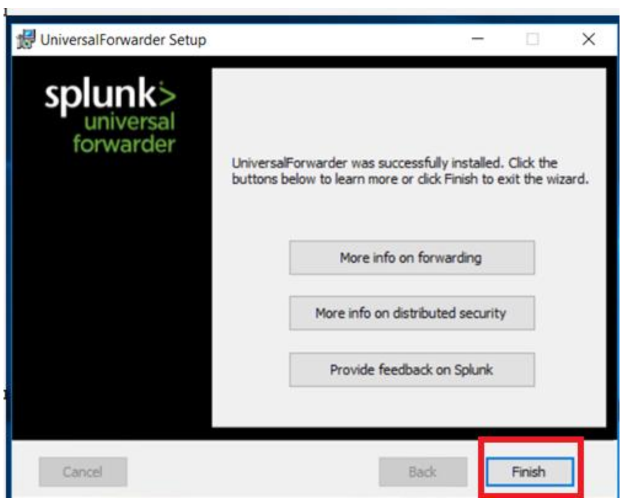
4. In Next step, IP Address of Splunk server, which is Kali Linux in this lab, is to be provided. As identified in step 4, IP address – 10.10.35.3 is provided and default port 8089 is provided.



5. In Next step, IP Address of Receiver Indexer, which is Kali Linux in this lab, is to be provided. As identified in step 4, IP address – 10.10.35.3 is provided and default port 9997 is provided.



6. Click on Next until installation is completed.

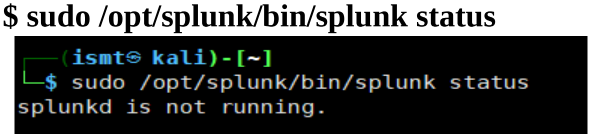


Kali Linux

Now, we need to add this windows server in Splunk for accepting logs.

1. Check the status of Splunk if it is running or not.

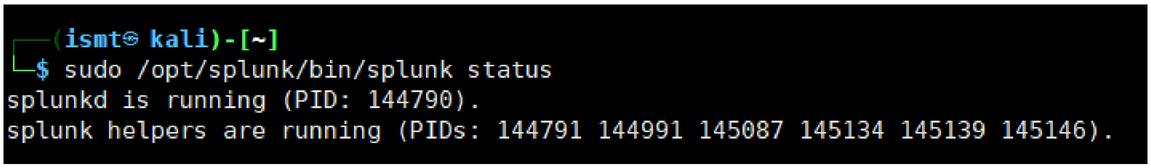
   $ sudo /opt/splunk/bin/splunk status



As shown above, splunk service is not running. Hence it needs to be started. If the service is in running state, no need to execute below command.

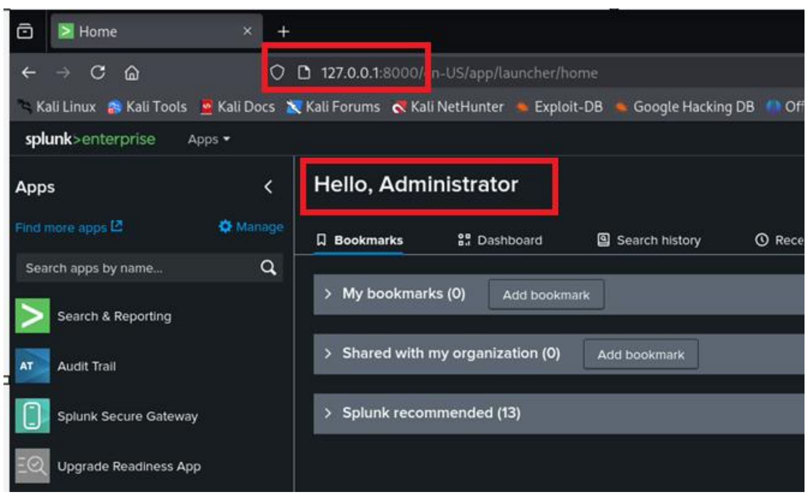
$ sudo /opt/splunk/bin/splunk start

$ sudo /opt/splunk/bin/splunk status

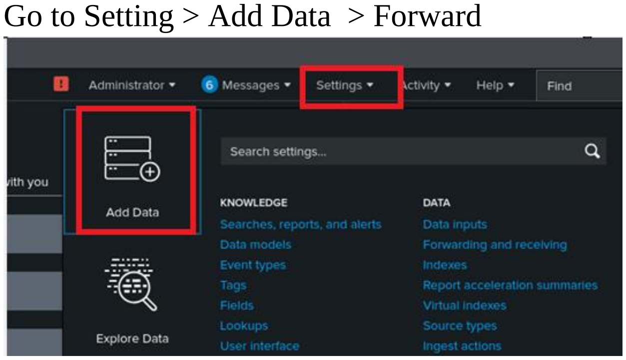


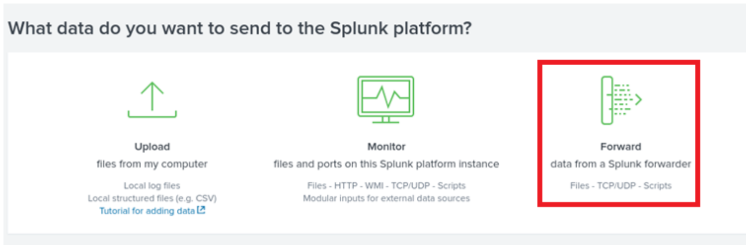
2. Access web portal for splunk, using username and password created above in step 6.

   http:// 127.0.0.1:8000



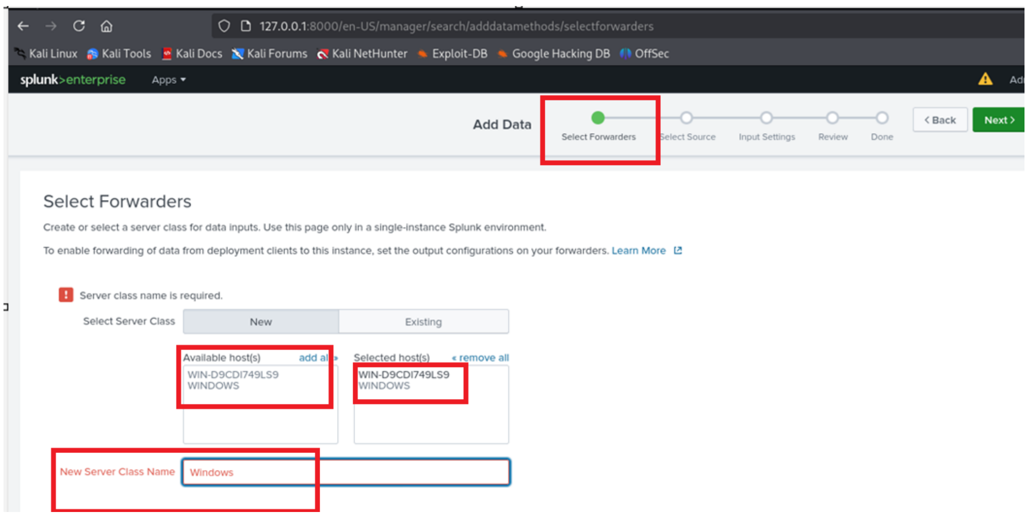
3. Go to Setting > Add Data > Forward



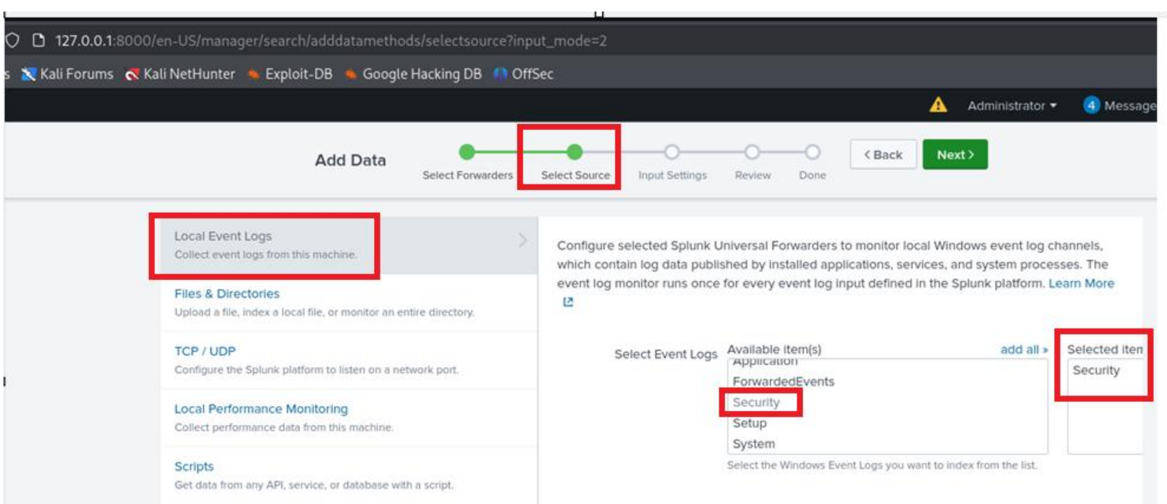


4. Click on available host to select as a forwarder, once selected it should be added to selected host section as well.

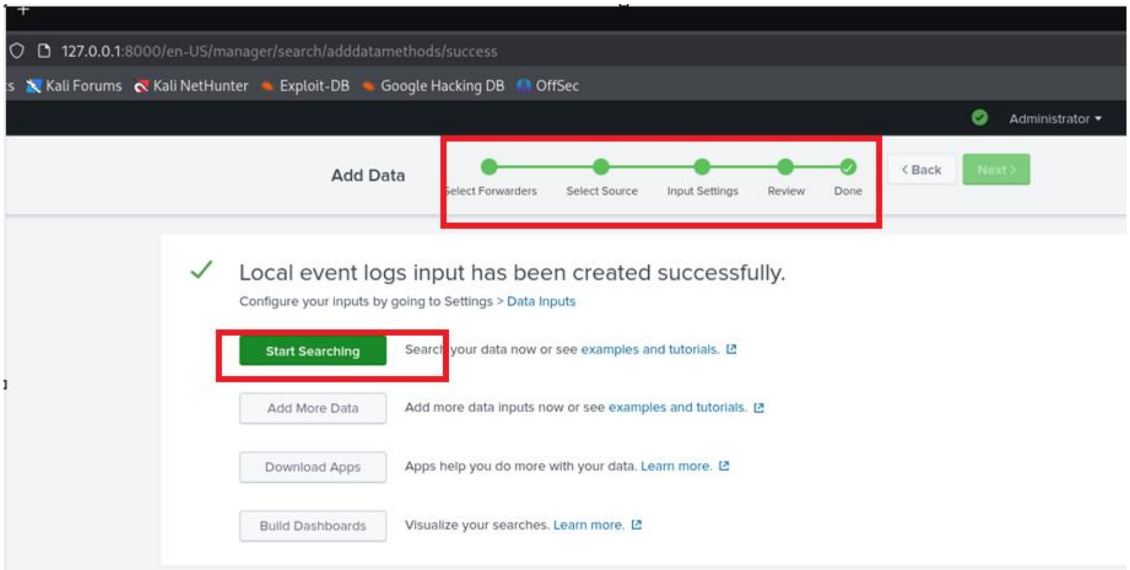
   Add New Server Class Name as "Windows". Finally click on Next



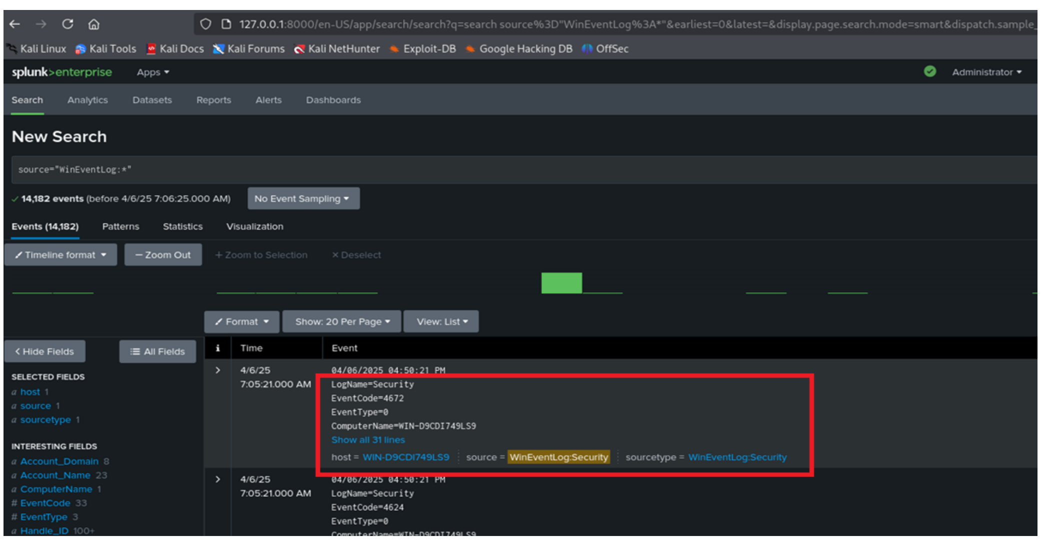
5. In select source option, select Local Event Logs and event type as security only. Then click on next.



6. Click next until configuration is completed and start searching for logs.



7. Finally we will see Security related event logs from Windows Server.



8. Let us test with a special use case.

Windows

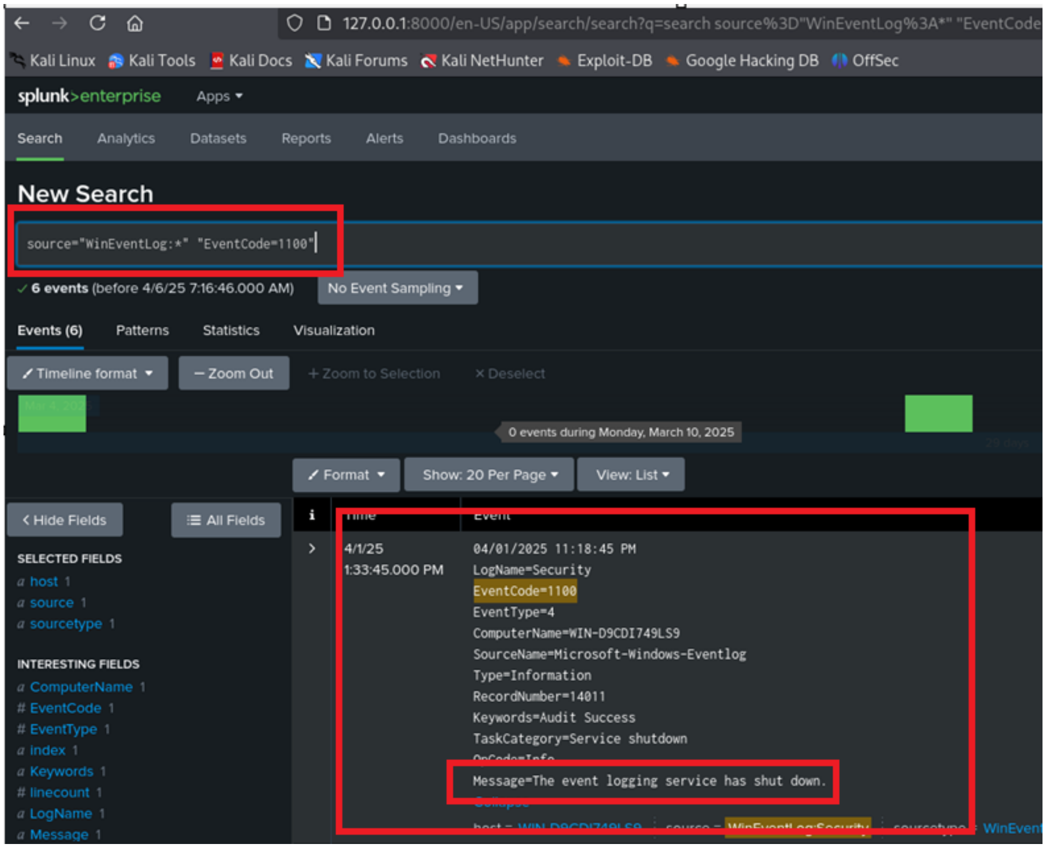
Restart (Reboot) windows server.

Kali Linux (Splunk)

Restarting event is captured with Windows Event Log ID – 1100. If we search for this specific id, information related to windows restart will be searched.

On Search Panel, search for:

source="WinEventLog:*" "EventCode=1100"



After Lab is completed, stop splunk service and revert the VM Network Adaptor to previous state as per screenshots in Virtual Box Step 1.

$ sudo /opt/splunk/bin/splunk stop